

WORLD LAYER MODEL — FORMAL PAPER SERIES
The Stability Mechanism of the Constant-Generation Chain:
From Zero-Origin Ground State to High-Dimensional Closure
A Structural Analysis of Mathematical Constants as
Generative Stabilizers

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Abstract

This paper formalizes the stability functions of the constant-generation chain within the World Layer Model (WLM) framework, treating each mathematical constant not as a numerical value discovered through measurement but as a structural stabilizer of a specific generative transition. We analyze the roles of 0 , 1 , π , e , φ , α , τ , Ω , c , and $\hbar$ across dimensional transitions D22–D27, establishing a unified model that connects recursive closure, self-amplification, self-similar expansion, dimensional coupling, full-cycle emergence, structural incompressibility, maximal deviation propagation, and minimal action within a single generative ontology. We provide formal definitions of the structural space $\mathcal{S}$, the offset operator Δ , the rotation and generative operators, and the dimensional domain maps. Proof sketches are given for the uniqueness of each constant as a stabilizer of its respective transition; the topological derivation of α is treated in a dedicated companion paper (Gu 2026a), and the Chaitin constant Ω is formally placed as an incompressibility boundary. The central claim is that positional order in the constant-generation chain is not conventional but structurally necessary.

Keywords: generative ontology; mathematical constants; dimensional closure; recursive stability; WLM framework; structural invariants; constant-generation chain; Kolmogorov complexity; fine-structure constant

Introduction

Mathematics has always harbored a peculiar tension between discovery and invention. Constants such as π , e , and φ are conventionally treated as facts about the mathematical universe — values that exist independently of any framework and are simply uncovered by analysis. What is rarely asked, however, is *why these particular values stabilize the processes they appear in*. Why should the ratio of a circle's circumference to its diameter be the same constant that governs oscillatory closure in differential equations? Why should the base of natural logarithms also be the unique fixed point of its own derivative? These are not coincidences of notation. They point, we argue, to a deeper structural logic in which each

constant is generated by — and in turn stabilizes — a particular type of recursive operation.

The World Layer Model (WLM) provides a framework within which this question becomes tractable. WLM, described in its foundational paper (Gu 2026b), organizes generative reality into 36 structural layers spanning dimensions D0 through D35 (including the Operator levels D28–D35). The central object is not a physical field but a generative sequence: beginning from the unbiased ground state 0, a minimal deviation — an *offset* — propagates through polarity, structure, and emergent world. Mathematical constants, in this reading, are the fixed points of the recursion archetypes that drive each transition. They are not discovered; they are what remains invariant when a generative process closes on itself.

This paper focuses on the dimensional range D22–D27, corresponding in the WLM architecture to the layers of Meaning, Meta-Meaning, Transcendence, Recursion, and Closure. It is precisely here that the constant-generation chain reaches its most explicit structural form. The transitions between these layers are governed by what we call *generative stabilizers*: constants whose values are not chosen but are the unique solutions to minimal-closure conditions imposed by the recursion structure itself.

The remainder of the paper is organized as follows. Section 2 provides the full formal apparatus: definitions of the structural space $\mathcal{S}$, operators, norms, and failure modes. Section 3 defines the dimensional domains D22–D27 and their transition maps. Sections 4 through 11 analyze each constant with proof sketches for uniqueness. The derivation of α is treated in a separate dedicated companion paper, which establishes its topological origin within the WLM framework (Gu 2026a); Section 7 of this paper accordingly provides only a structural summary and a pointer to that external derivation. Section 12 assembles the unified chain model. Section 13 addresses limitations and scope. Section 14 concludes. An appendix collects all mathematical preliminaries needed for self-contained reading.

Formal Mathematical Foundations

This section establishes the formal apparatus that underpins the remainder of the paper. The definitions here are not ad hoc; they are the minimal scaffolding required to give precise meaning to the claims in Sections 4–11.

The Structural Space $\mathcal{S}$

Definition 1 (Structural Space). *The structural space $\mathcal{S}$ is a complete, separable metric space equipped with a partial order $\preceq$ and a distinguished element $\mathbf{0} \in \mathcal{S}$, satisfying:*

1. **Completeness:** Every Cauchy sequence in $\mathcal{S}$ converges in $\mathcal{S}$.
2. **Separability:** $\mathcal{S}$ admits a countable dense subset.
3. **Ground element:** $\mathbf{0}$ is the unique minimal element under $\preceq$; that is, $\mathbf{0} \preceq s$ for all $s \in \mathcal{S}$.

4. **Generative closure:** $\mathcal{S}$ is closed under a family of generative operators $\{\mathcal{F}_k\}_{k \geq 0}$ (defined below).

Concretely, $\mathcal{S}$ may be realized as a Hilbert space H of square-summable sequences over a countable alphabet of structural primitives, with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|s\| = \sqrt{\langle s, s \rangle}$. The ordering $\preceq$ is the standard ℓ^2 cone ordering.

The choice of a Hilbert-space realization is not essential to the framework; the key properties are completeness and the existence of a distinguished zero element. Alternative realizations in Banach spaces or in the category of measurable spaces are structurally equivalent for the purposes of this paper.

The Structural Norm $\|\cdot\|$

Definition 2 (Structural Norm). The structural norm $\|\cdot\|: \mathcal{S} \rightarrow \mathbb{R}_{\geq 0}$ is the norm induced by

the inner product on the Hilbert-space realization of $\mathcal{S}$: $\|s\| = \sqrt{\sum_{i=0}^{\infty} s_i^2}$, $s =$

$(s_0, s_1, s_2, \dots) \in \ell^2$. This norm satisfies:

1. $\|s\| = 0 \Leftrightarrow s = \mathbf{0}$.
2. $\|\lambda s\| = |\lambda| \|s\|$ for all $\lambda \in \mathbb{R}$.
3. $\|s + t\| \leq \|s\| + \|t\|$ (triangle inequality).

The structural distance between two configurations is $d(s, t) = \|s - t\|$.

The structural norm measures the total deviation of a configuration from the ground state. It will appear in the definition of the offset operator and in the minimality conditions for each constant.

The Offset Operator Δ

Definition 3 (Offset Operator). The offset operator $\Delta: \mathcal{S} \rightarrow \mathcal{S}$ measures the deviation of a structural configuration from the ground state $\mathbf{0}$. It is defined by: $\Delta(s) = s - \mathbf{0} = s$, with the associated offset magnitude: $|\Delta|(s) = \|s\|$. A configuration $s \in \mathcal{S}$ is called unbiased if $\Delta(s) = \mathbf{0}$, i.e., $s = \mathbf{0}$. It is called offset if $\Delta(s) \neq \mathbf{0}$.

In a dynamical context, the offset operator generalizes to a time-dependent version. Let $\mathcal{S}^{\mathbb{R}} = \{s: \mathbb{R} \rightarrow \mathcal{S}\}$ denote the space of structural trajectories. The differential offset is:

$$\dot{\Delta}(s)(t) = \frac{d}{dt} s(t),$$

which measures the rate at which a configuration departs from stationarity. The generative dynamics of the WLM chain are governed by the condition that $\dot{\Delta}$ remains bounded and converges to zero at stable fixed points.

The Rotation Operator $\mathcal{R}_\theta$

Definition 4 (Rotation Operator). For $\theta \in [0, 2\pi)$, the rotation operator $\mathcal{R}_\theta: \mathcal{S} \rightarrow \mathcal{S}$ is a one-parameter family of unitary operators on the Hilbert-space realization of $\mathcal{S}$, satisfying:

1. $\mathcal{R}_0 = \text{id}_{\mathcal{S}}$ (identity at zero angle).
2. $\mathcal{R}_{\theta_1} \circ \mathcal{R}_{\theta_2} = \mathcal{R}_{\theta_1 + \theta_2 \bmod 2\pi}$ (group law).
3. $\|\mathcal{R}_\theta(s)\| = \|s\|$ for all $s \in \mathcal{S}$ (norm preservation).
4. $\mathcal{R}_\theta(\mathbf{0}) = \mathbf{0}$ (ground state invariance).

Concretely, on a two-dimensional subspace spanned by basis vectors $e_1, e_2 \in \mathcal{S}$, the operator $\mathcal{R}_\theta$ acts as the standard rotation matrix: $\mathcal{R}_\theta \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}$. On higher-dimensional subspaces, $\mathcal{R}_\theta$ is extended by block-diagonalization.

The Generative Operator $\mathcal{F}$

Definition 5 (Generative Operator). The generative operator $\mathcal{F}: \mathcal{S} \rightarrow \mathcal{S}$ is a nonlinear, contractive map satisfying:

1. **Fixed point:** $\mathcal{F}(\mathbf{0}) = \mathbf{0}$ (the ground state is preserved).
2. **Contractivity near 0:** There exists $L < 1$ such that $\|\mathcal{F}(s) - \mathcal{F}(t)\| \leq L\|s - t\|$ for all s, t in a neighborhood of $\mathbf{0}$.
3. **Offset amplification away from 0:** For $\|s\|$ above a threshold $\eta > 0$, $\|\mathcal{F}(s)\| > \|s\|$, representing the generative expansion of the dimensional chain.

The family $\{\mathcal{F}_k\}_{k=0}^{35}$ denotes the dimensional-specific generative operators, where $\mathcal{F}_k: \mathcal{S}_k \rightarrow \mathcal{S}_{k+1}$ maps configurations in dimension k to configurations in dimension $k + 1$.

The tension between properties (ii) and (iii) is the engine of the generative chain: near the ground state, the operator is contractive and stable; away from it, it is expansive and generative. The constants analyzed in this paper are precisely the values at which this tension is resolved in a dimensionally consistent manner.

Formal Definitions of Generative Failure Modes

Before analyzing each constant, we formalize the failure modes they prevent.

Definition 6 (Divergence). A trajectory $s: \mathbb{R}_{\geq 0} \rightarrow \mathcal{S}$ is said to diverge if: $\lim_{t \rightarrow \infty} \|s(t)\| = +\infty$. In dimensional terms, divergence at transition $\mathcal{F}_k$ means that the image $\mathcal{F}_k(\mathcal{S}_k)$ is unbounded in $\mathcal{S}_{k+1}$, preventing the formation of a stable dimensional domain.

Definition 7 (Collapse). A trajectory $s: \mathbb{R}_{\geq 0} \rightarrow \mathcal{S}$ collapses if: $\lim_{t \rightarrow \infty} \|s(t)\| = 0$, i.e., it converges to the ground state $\mathbf{0}$. At a dimensional transition, collapse means the image $\mathcal{F}_k(\mathcal{S}_k)$ degenerates to $\{\mathbf{0}\}$, annihilating the generative content of dimension k .

Definition 8 (Desynchronization). *Two trajectories $s_1, s_2: \mathbb{R}_{\geq 0} \rightarrow \mathcal{S}$ are desynchronized if there exists no $T > 0$ such that $s_1(t + T) = s_2(t)$ for all t . In the context of the generative chain, desynchronization at transition $\mathcal{F}_k$ means that the periodic structure of $\mathcal{S}_k$ does not lift coherently to $\mathcal{S}_{k+1}$: the cycles of adjacent dimensions become incommensurate.*

Definition 9 (Incompressibility Overflow). *Given a structural configuration $s \in \mathcal{S}$ and a compression map $\mathcal{C}: \mathcal{S} \rightarrow \{0,1\}^*$, the structural density of s is: $\rho(s) = \lim_{n \rightarrow \infty} \frac{K(s \upharpoonright_n)}{n}$, where K denotes Kolmogorov complexity and $s \upharpoonright_n$ is the n -bit prefix of s . Incompressibility overflow at transition $\mathcal{F}_k$ occurs when the structural density of elements in $\mathcal{F}_k(\mathcal{S}_k)$ exceeds 1, which is impossible by definition but is approached asymptotically when the compression requirements of $\mathcal{S}_{k+1}$ exceed the information capacity of the chain up to dimension k .*

Dimensional Domains D22–D27: Formal Specification

Layer Ontology and Mathematical Inhabitants

Each dimensional domain $\mathcal{S}_k$ is a closed subspace of $\mathcal{S}$ with specific structural properties. The following table summarizes the mathematical objects inhabiting each layer in the range D22–D27:

Layer	WLM Name	Mathematical Objects
D22	Meaning	Closed semantic graphs; finitely generated groups
D23	Meta-Meaning	Self-referential structures; sheaves over D22
D24	Transcendence	Periodic orbits; compact Lie group actions
D25	Recursion	Analytic semigroups; exponential maps
D26	Coherence/Rendering	Fiber bundles over D25; coupled field theories
D27	Closure	Maximal-density configurations; complete metric spaces

More precisely, each $\mathcal{S}_k$ is defined inductively:

Definition 10 (Dimensional Domain). *For $k \in \{22, 23, 24, 25, 26, 27\}$, the dimensional domain $\mathcal{S}_k$ is the unique (up to isomorphism) closed subspace of $\mathcal{S}$ satisfying: $\mathcal{S}_k = \text{Fix}(\mathcal{F}_{k-1}) \cap \ker(\mathcal{F}_k - \lambda_k \cdot \text{id})$, where $\text{Fix}(\mathcal{F}_{k-1})$ denotes the fixed-point set of the generative operator at level $k - 1$, and λ_k is the spectral eigenvalue determined by the stabilizing constant C_k (see Table 1 in Section 12).*

The intuition behind (3.1) is that $\mathcal{S}_k$ consists precisely of those configurations that are stable under the preceding operator $\mathcal{F}_{k-1}$ (they have “settled” into dimension k) while remaining sensitive to the next operator $\mathcal{F}_k$ (they have not yet closed into dimension $k + 1$).

Transition Maps and Stability Conditions

Definition 11 (Dimensional Transition Map). *The transition map $T_k: \mathcal{S}_k \rightarrow \mathcal{S}_{k+1}$ is defined as the restriction of $\mathcal{F}_k$ to $\mathcal{S}_k$: $T_k = \mathcal{F}_k|_{\mathcal{S}_k}$. A constant C_k stabilizes the transition T_k if and only if the following three conditions hold simultaneously:*

1. **Non-divergence:** $\sup_{s \in \mathcal{S}_k, \|s\| \leq 1} \|T_k(s)\| < \infty$.
2. **Non-collapse:** $\inf_{s \in \mathcal{S}_k, \|s\|=1} \|T_k(s)\| > 0$.
3. **Periodicity/closure:** *The orbit $\{T_k^n(s)\}_{n \geq 0}$ is pre-compact in $\mathcal{S}_{k+1}$ for all $s \in \mathcal{S}_k$.*

The requirement that all three conditions hold simultaneously is what makes the stabilizing constant unique in each case. Condition (i) prevents divergence, condition (ii) prevents collapse, and condition (iii) ensures that the transition produces stable structure in the target domain rather than an aperiodic orbit.

Why Stabilizers Are Necessary

The need for explicit stabilizers at each dimensional transition follows from a general instability theorem, which we state informally here.

Theorem 1 (Generic Instability of Dimensional Transitions). *For a generic operator $\mathcal{F}_k$ (in the sense of Baire category in the space of bounded operators on $\mathcal{S}$), at least one of the three conditions in Definition 11 fails. In other words, the set of operators that simultaneously satisfy non-divergence, non-collapse, and pre-compactness is a measure-zero subset of the space of all operators $\mathcal{S}_k \rightarrow \mathcal{S}_{k+1}$.*

This theorem, which follows from standard arguments in functional analysis (see, e.g., Kato 1966 on perturbation theory for linear operators), implies that stable dimensional transitions are non-generic. The constants π , e , φ , α , Ω are precisely the selection devices that pick out the non-generic, stable transitions from the generic, unstable background.

π : Recursive Closure at $D_{23} \rightarrow D_{24}$

Once the identity offset is established and the self-similar expansion of φ has populated $\mathcal{S}_{23}$, the immediate danger is divergence: a recursion that applies the offset operator indefinitely without ever returning to a closed configuration. The constant π is precisely the stabilizer that prevents this.

Within the WLM framework, π is not primarily a geometric constant. It is the *minimal return ratio* of a closed recursive loop: the unique value such that a generative process that applies an angular deviation of π per half-cycle returns exactly to its starting configuration after one full cycle.

Proposition 1 (Uniqueness of π as Closure Constant). *Let $\theta > 0$ be such that $\mathcal{R}_\theta$ satisfies all three stability conditions of Definition 11 for the transition $T_{23}: \mathcal{S}_{23} \rightarrow \mathcal{S}_{24}$. Then $\theta = \pi$ (modulo 2π).*

Proof sketch. Condition (iii) of Definition 11 requires that the orbit $\{\mathcal{R}_\theta^n(s)\}_{n \geq 0}$ be pre-compact. By the spectral theorem for unitary operators on Hilbert spaces, the spectrum of $\mathcal{R}_\theta$ consists of complex numbers $e^{in\theta}$. Pre-compactness of orbits requires that the spectrum be confined to the unit circle, which is satisfied for all θ . However, condition (ii) additionally requires that the orbit does not collapse, i.e., that the only fixed point of $\mathcal{R}_\theta$ in $\mathcal{S}_{23}$ is $\mathbf{0}$. This eliminates $\theta = 0$ and, by periodicity arguments, all rational multiples of 2π with denominator less than 3. The minimal remaining value compatible with conditions (i)–(iii) and with the self-similar structure of $\mathcal{S}_{23}$ (inherited from the φ -stabilization at T_{22}) is $\theta = \pi$.

The uniqueness follows from the observation that any $\theta \in (\pi, 2\pi)$ would produce an orbit that over-rotates before closure, generating a gap in the periodic structure of $\mathcal{S}_{24}$ that is incompatible with the self-similar tiling inherited from $\mathcal{S}_{23}$. ▀

Definition 12 (Closure Condition for π). The π -closure condition at D24 is: $\mathcal{R}_{2\pi}(s) = s \quad \forall s \in \mathcal{S}_{24}$, or equivalently, the orbit of any $s \neq \mathbf{0}$ under $\mathcal{R}_\theta$ has minimal period 2π . The averaging identity: $\frac{1}{2\pi} \int_0^{2\pi} \mathcal{R}_\theta(s) d\theta = \mathbf{0} \quad \forall s \perp \mathbf{0}$ confirms that π -closure produces a domain of zero mean deviation, consistent with the fixed-point structure of $\mathcal{S}_{24}$.

What would happen with a different constant? Suppose one replaced π with $\pi' = 3$ (approximately π but rational). Then $\mathcal{R}_{2\pi'}$ would close after a rational fraction of the circle, producing a periodic lattice rather than a continuous closure. The dimensional domain $\mathcal{S}_{24}$ would be discretized in a way that is incompatible with the continuous analytic semigroup structure required at $\mathcal{S}_{25}$, where e takes over as stabilizer. This is the concrete failure mode that π -closure prevents.

***e*: Self-Amplifying Stability at D24 → D25**

Definition 13 (Self-Amplifying Stability). A generative process $\mathcal{E}: \mathbb{R}_{\geq 0} \rightarrow \mathcal{S}$ is called self-amplifying stable if: $\frac{d}{dt} \mathcal{E}(t) = \lambda \cdot \mathcal{E}(t)$, $\mathcal{E}(0) = \mathbf{1}$, for some $\lambda \in \mathbb{R}$, and if $\mathcal{E}$ remains bounded on compact intervals while remaining bounded away from $\mathbf{0}$ for all finite t .

Proposition 2 (Uniqueness of e as Self-Amplification Base). The unique base $b > 1$ for which the self-amplifying stable process $\mathcal{E}(t) = b^t$ satisfies the transition stability conditions (Definition 11) for $T_{24}: \mathcal{S}_{24} \rightarrow \mathcal{S}_{25}$, and for which the generator of the associated semigroup is equal to its own eigenfunction, is $b = e$.

Proof sketch. The transition T_{24} maps the compact, π -closed domain $\mathcal{S}_{24}$ into the analytically expanding domain $\mathcal{S}_{25}$. Condition (i) (non-divergence on compact intervals) is satisfied for any $b < \infty$. Condition (ii) (non-collapse) requires $b > 1$. The additional structural requirement — that the generator of the associated one-parameter semigroup $\{e^{tA}\}_{t \geq 0}$ commutes with the spectral projections of $\mathcal{S}_{25}$ — imposes the constraint: $\frac{d}{dt} \big|_{t=0} b^t = \ln b = 1 \Rightarrow b = e$. Any other base $b \neq e$ produces a generator with $\ln b \neq 1$, breaking the self-referential eigenfunction property and causing the semigroup to fail condition (iii) (pre-compactness of orbits in $\mathcal{S}_{25}$). ▀

The key equation:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

encodes the fact that e -governed growth is the continuum limit of discrete compounding, precisely the structure of the D24 $\rightarrow$ D25 transition where the discrete periodic domain of D24 gives way to continuous analytic expansion.

φ : Self-Similar Expansion at D22 $\rightarrow$ D23

The golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ governs the earlier transition D22 $\rightarrow$ D23, corresponding to the shift from Meaning to Meta-Meaning.

Proposition 3 (Uniqueness of φ as Self-Similarity Constant). *The unique scaling ratio $\lambda > 1$ such that the transition $T_{22}: \mathcal{S}_{22} \rightarrow \mathcal{S}_{23}$ satisfies all three stability conditions and produces a self-similar tiling of $\mathcal{S}_{23}$ by copies of $\mathcal{S}_{22}$ without overlap or gap is $\lambda = \varphi$.*

Proof sketch. Self-similar tiling requires: $\mathcal{S}_{23} \cong \mathcal{S}_{22} \sqcup \lambda^{-1}\mathcal{S}_{23}$, which gives the fixed-point equation $\lambda^2 - \lambda - 1 = 0$, with positive solution $\lambda = \varphi = \frac{1+\sqrt{5}}{2}$.

Any $\lambda < \varphi$ produces gaps in the tiling (structural incompleteness at D23); any $\lambda > \varphi$ produces overlaps (structural conflict). The uniqueness of φ follows from the uniqueness of the positive root of $\lambda^2 = \lambda + 1$. ▫

Definition 14 (Self-Similarity Fixed-Point Equation). *The φ -self-similarity condition is the fixed-point equation: $\varphi = 1 + \frac{1}{\varphi}$, equivalently $\varphi^2 = \varphi + 1$. The continued fraction*

representation: $\varphi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{\ddots}}}$ establishes φ as the most irrational number (slowest rational

convergence), making proportional structures governed by φ maximally resistant to resonant collapse at D22.

α : Dimensional Coupling at D25 $\rightarrow$ D26

The fine-structure constant $\alpha \approx 1/137^1$ occupies a structurally singular position in the chain: it is the unique coupling constant mediating the transition between the generative layer $\mathcal{S}_{25}$ and the rendering layer $\mathcal{S}_{26}$. The full topological derivation of $\alpha \approx 1/137$ is presented in the companion paper *The Structural Origin of the Fine-Structure Constant: A Topological Derivation of 1/137 in WLM* (Gu 2026a). In this paper we summarize only the structural role of α as the unique dimensional coupling constant stabilizing the D25 $\rightarrow$ D26 transition, and refer the reader to that paper for all geometric and index-theoretic details.

¹ Full derivation in Gu (2026a).

Structural Role of α in the Chain

Within the constant-generation chain, α stabilizes the coupling between the generative layer (D25) and the rendering layer (D26). It ensures that the projection from the exponential generative domain into the appearance domain is neither too strong (which would cause collapse of the generative layer into the rendering layer, destroying the distinction between structure and appearance) nor too weak (which would leave the rendering layer decoupled from its generative source, producing incoherent appearances). Its numerical value, $\alpha \approx 1/137$, is not introduced empirically but arises from a topological congruence system derived in the companion paper.

The transition $T_{25}: \mathcal{S}_{25} \rightarrow \mathcal{S}_{26}$ is the most structurally sensitive in the chain because it crosses the boundary between the purely generative domain (where structures exist as potential configurations) and the rendering domain (where structures become observable). No other constant in $\mathcal{C}$ can perform this function: constants governing closure (π, τ) operate on the wrong spectral structure; constants governing growth (e) do not provide a coupling ratio; and the incompressibility constant (Ω) operates at a different density regime.

Result from the Companion Paper

The companion paper (Gu 2026a) establishes, via a system of three geometric congruence conditions (derived respectively from the Gysin sequence, normal bundle spin structure, and the Atiyah-Singer index theorem on a twisted Dirac operator) combined via the Chinese Remainder Theorem, that the topological coupling invariant $c_1(P)$ of the relevant $U(1)$ -bundle over the WLM closure manifold satisfies:

$$c_1(P) \equiv 137 \pmod{300}.$$

The leading residue 137 is identified with α^{-1} at leading order, with sub-leading corrections of order $O(\alpha/\pi)$ matching the known Schwinger expansion. The geometric congruence system (mod 5, mod 20, mod 15) and its CRT resolution to (7.1) are fully derived in that companion paper; here we treat α only as the resulting stabilizer constant.

Proposition 4 (Uniqueness of α as Coupling Stabilizer). *α is the unique constant satisfying all three stability conditions of Definition 11 for the transition $T_{25}: \mathcal{S}_{25} \rightarrow \mathcal{S}_{26}$. A full proof of uniqueness is provided in the companion paper (Gu 2026a); here we state the result without reproducing the derivation.*

The stability condition for the D25 $\rightarrow$ D26 transition takes the form:

$$\int_{\mathcal{S}_{26}} \mathcal{K}(s, s') d\mu(s') = \alpha \cdot s \quad \forall s \in \mathcal{S}_{25},$$

where $\mathcal{K}$ is the cross-dimensional coupling kernel and μ is the natural measure on $\mathcal{S}_{26}$. Equation (7.2) states that the projection of each generative structure into the rendering layer is proportional to the structure itself, with proportionality constant α , establishing D26 as a coupled but not collapsed domain.

τ : Full-Cycle Emergence at D24 $\rightarrow$ D25

Definition 15 (Full-Cycle Emergence Condition). *A recursion $\mathcal{R}$ on $\mathcal{S}_{24}$ satisfies the full-cycle emergence condition if: $\mathcal{R}^{(\tau)} = \text{id}_{\mathcal{S}_{24}}$, and $\mathcal{R}^{(\theta)} \neq \text{id}_{\mathcal{S}_{24}}$ for all $0 < \theta < \tau$.*

The constant $\tau = 2\pi$ is the minimal period satisfying (8.1) in $\mathcal{S}_{24}$. This can be seen from the spectral decomposition of $\mathcal{R}_\theta$: the eigenvalues of $\mathcal{R}_\theta$ are $\{e^{in\theta} : n \in \mathbb{Z}\}$, and the condition $\mathcal{R}_\tau = \text{id}$ requires $e^{in\tau} = 1$ for all n , i.e., $\tau \in 2\pi\mathbb{Z}$. The minimal positive solution is $\tau = 2\pi$.

The distinction from π is structural: π is the half-period (the angle at which the orbit is reversed), while τ is the full period (the angle at which the orbit returns to the origin having traversed its complete generative range). A recursion closed at π generates only half the available structural content of $\mathcal{S}_{24}$; the full content requires a τ -cycle.

Ω : Incompressibility at D26 $\rightarrow$ D27 — Formal Treatment

Structural Density and the Chaitin Constant

Definition 16 (Structural Density Measure). *Let $\mu: \mathcal{S} \rightarrow [0,1]$ be the structural density measure defined by: $\mu(s) = \lim_{n \rightarrow \infty} \frac{K(s \upharpoonright_n)}{n}$, where K is Kolmogorov complexity (with respect to a fixed universal Turing machine U) and $s \upharpoonright_n$ is the projection of s onto the first n coordinates in the Hilbert-space basis. By the law of large numbers for Kolmogorov complexity (Li and Vitányi 2008), $\mu(s)$ exists and equals 1 almost surely (with respect to the uniform measure on $\{0,1\}^{\mathbb{N}}$).*

Definition 17 (Chaitin Constant as Structural Boundary). *The Chaitin halting probability: $\Omega_U = \sum_{p: U(p) \text{ halts}} 2^{-|p|}$ is a real number in $(0,1)$ that is maximally incompressible: $K(\Omega_U \upharpoonright_n) \geq n - O(1)$, i.e., its structural density is $\mu(\Omega_U) = 1$.*

Formal Placement at D26 $\rightarrow$ D27

The transition $T_{26}: \mathcal{S}_{26} \rightarrow \mathcal{S}_{27}$ is the transition from the coupled rendering layer to the maximal-density closure domain. By definition (def:Dk), $\mathcal{S}_{27}$ consists of configurations with maximal structural density $\mu(s) = 1$. The question is: what constant stabilizes the transition into this domain?

Proposition 5 (Uniqueness of Ω as Incompressibility Stabilizer). *The unique constant that stabilizes T_{26} without causing incompressibility overflow (Definition 9) is the Chaitin constant Ω .*

Proof sketch. Any constant with $\mu(C) < 1$ would, by the density amplification property of T_{26} , map configurations in $\mathcal{S}_{26}$ to configurations in $\mathcal{S}_{27}$ with density exceeding $\mu(C)$ — in the limit, exceeding 1, which is impossible. Only a constant with $\mu(C) = 1$ can serve as the “ceiling” that T_{26} approaches without overflow.

Among all constants with $\mu(C) = 1$, Ω is the canonical choice because it encodes the incompressibility of the halting problem itself, which is the archetypal incompressible decision problem. Any other constant with $\mu(C) = 1$ would be Turing-equivalent to Ω (by the invariance theorem for Kolmogorov complexity), and would serve the same stabilizing function up to a constant additive factor. ▫

Connection to Gödel and Structural Completeness

The Chaitin constant is related to the incompleteness theorems through the following chain: Gödel (1931) showed that any sufficiently powerful formal system contains true but unprovable statements. Turing (1936) showed that the halting problem is undecidable. Chaitin (1975) showed that Ω encodes the information-theoretic content of both results. In the WLM framework, this means that the closure of $\mathcal{S}_{27}$ — the final closed domain of the dimensional chain — necessarily contains irreducible information. A system that achieved complete self-description through compression alone would contradict the incompressibility of Ω . The placement of Ω at D27 is thus not merely convenient; it is the structural expression of Gödelian incompleteness within the generative chain.

The Ground State and Identity Offset

Definition 18 (Zero-Origin Ground State). *The zero-origin ground state $\mathcal{G}_0$ is the unique element $\mathbf{0} \in \mathcal{S}$ satisfying $\mathcal{F}(\mathbf{0}) = \mathbf{0}$ and $\|\mathbf{0}\| = 0$.*

Definition 19 (Identity Offset Unit). *The identity offset unit $\mathbf{1}$ is defined as: $\mathbf{1} := \min\{\|\Delta(s)\| \mid s \neq \mathcal{G}_0, s \in \mathcal{S}\}$, the smallest non-zero deviation sustainable against the contractive action of $\mathcal{F}$ near $\mathbf{0}$. The structural space decomposes as: $\mathcal{S} = \{\mathcal{G}_0 + n \cdot \mathbf{1} \mid n \in \mathbb{Z}_{\geq 0}\} \cup \mathcal{S}_{\text{cont}}$, where $\mathcal{S}_{\text{cont}}$ is the space of continuous (non-discrete) deformations.*

c and $\hbar$: Global Propagation Bounds

c : Maximum Deviation Propagation Speed

Definition 20 (Maximum Propagation Speed). *The propagation speed of a deviation $\delta: \mathcal{S} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is: $v_{\text{prop}} = \sup \left\{ \frac{\|x - x_0\|}{|t - t_0|} \mid \delta(x_0, t_0) \neq 0, \delta(x, t) \neq 0, t > t_0 \right\}$. The constant c is the supremum of v_{prop} over all possible deviation fields δ .*

The finiteness of c follows from the requirement that the dimensional transitions $\mathcal{F}_k$ have bounded spectral radius. If deviations could propagate at infinite speed, the sequential structure of the generative chain — each layer depending causally on the preceding one — would collapse. The chain's layered architecture is thus the structural origin of the light-speed bound.

$\hbar$: Global Minimal Action

Definition 21 (Minimal Generative Action). *The minimal action associated with a single generative transition in the rendering layer is: $\mathcal{A}_{\min} = \hbar$, meaning that no generative event can occur with action $E \cdot \Delta t < \hbar/2$. This is the energy-time uncertainty bound, here derived from the spectral gap of the generative operator: $\sigma_{\min}(\mathcal{F}_k) \cdot \Delta t \geq \hbar/2$, where $\sigma_{\min}(\mathcal{F}_k)$ is the minimal non-zero eigenvalue of $\mathcal{F}_k$ in the energy representation.*

Together, c and $\hbar$ define a two-sided bound: c is a global upper bound on propagation, and $\hbar$ is a global lower bound on action. They do not govern any single dimensional transition but constrain the operator norm and spectral gap of every $\mathcal{F}_k$ simultaneously.

The Unified Model: Formal Assembly

The Constant-Generation Chain as a Formal Sequence

Definition 22 (Constant-Generation Chain). *The constant-generation chain is the ordered sequence: $\mathcal{C} = (C_0, C_1, C_2, C_3, C_4, C_5, C_6, C_7, C_8, C_9) = (0, 1, \varphi, \pi, \tau, e, \alpha, \Omega, c, \hbar)$, together with the mapping: $C_k \mapsto Dk$, $k = 0, 1, 2, \dots$ according to Table 1 below.*

Table 1. The constant-generation chain.

Index	Constant	Generative Role	Transition / Scope
C_0	0	Zero-origin ground state	Reference frame
C_1	1	Identity offset unit	Minimal stable distinction
C_2	φ	Self-similar expansion	D22 $\rightarrow$ D23
C_3	π	Recursive closure	D23 $\rightarrow$ D24
C_4	τ	Full-cycle emergence	D24 internal
C_5	e	Self-amplifying stability	D24 $\rightarrow$ D25
C_6	α	Dimensional coupling (derivation: Gu 2026a)	D25 $\rightarrow$ D26
C_7	Ω	Incompressibility boundary	D26 $\rightarrow$ D27
C_8	c	Maximum propagation speed	Global
C_9	$\hbar$	Minimal generative action	Global

Dependency Graph

The structural dependencies between constants are as follows. Each constant in the chain presupposes all constants that precede it:

An arrow $A \rightarrow B$ means: the stabilizing function of B is only well-defined once the transition governed by A has been completed. Replacing any node in this graph with a structurally incompatible constant would propagate failure to all downstream nodes.

Minimality and Completeness

Theorem 2 (Positional Necessity of the Constant-Generation Chain). *The constant-generation chain $\mathcal{C}$ defined in (12.1) is:*

1. **Minimal:** No proper subset of $\mathcal{C}$ simultaneously stabilizes all transitions $T_{22}, \dots, T_{26}$ and satisfies the global bounds on propagation and action.
2. **Complete:** Every transition T_k for $k \in \{22, 23, 24, 25, 26\}$ has a unique stabilizer in $\mathcal{C}$.
3. **Positionally necessary:** Each C_k occupies the unique position in $\mathcal{C}$ where it is structurally capable of performing its stabilizing function.

Proof outline.

- *Minimality:* By Theorem 1, each transition T_k is generically unstable. The chain is minimal because each constant is the unique solution to the closure condition of its respective transition (Propositions 1, 2, 3, 5). Removing any constant leaves its transition unstabilized.
- *Completeness:* Each transition T_k has exactly one stabilizer in $\mathcal{C}$ by the uniqueness arguments in Sections 4–9.
- *Positional necessity:* Suppose C_j is moved to position $i \neq j$. Then the transition T_{i-1} receives a stabilizer with the wrong structural properties (wrong fixed-point type, wrong spectral gap, or wrong incompressibility density), and by the dependency graph above, all downstream transitions also fail. ▫

Limitations and Scope

Academic honesty requires explicit statement of what this framework does and does not claim.

What the Model Does Not Claim

The WLM constant-generation chain does not claim to *derive* the numerical values of e , π , φ , c , and $\hbar$ from purely combinatorial data — these values are fixed by existing mathematics and physics, and the WLM framework *interprets* their structural roles rather than computing them anew. The case of α is different: its topological derivation from geometric congruence conditions is the subject of a dedicated companion paper (Gu 2026a), which establishes $c_1(P) \equiv 137 \pmod{300}$ and its connection to the measured value $\alpha^{-1} \approx 137.036$. Whether the full renormalization argument reproduces that value to all orders remains an open problem addressed in that paper.

The framework also does not claim that the dimensional numbering (D22–D27) is uniquely determined by the constants. The assignment of φ to D22 rather than to some other dimension reflects the internal logic of the WLM layer architecture; a reader who does not accept that architecture must treat the dimensional assignments as additional postulates.

Assumptions the Model Depends On

The argument rests on four main assumptions:

1. The structural space $\mathcal{S}$ is a Hilbert space (or can be realized as one without loss of generality for the properties analyzed here).
2. The generative operators $\mathcal{F}_k$ are bounded and have well-defined spectra.
3. The dimensional domains $\mathcal{S}_k$ are well-defined closed subspaces satisfying Definition 10.
4. The self-similar tiling condition (6.1) correctly models the $D22 \rightarrow D23$ transition.

Relaxing assumption (i) to Banach spaces would require replacing the spectral theorem with more general operator theory; the qualitative conclusions are expected to be unchanged, but the proofs would require modification.

Empirical Predictions and Falsifiability

The framework makes one concrete empirical prediction, articulated fully in the companion paper (Gu 2026a): the geometric derivation of α should, upon completion of the full renormalization argument, reproduce the measured value $\alpha^{-1} = 137.035999 \dots$ without additional fitting parameters. If it does not, the placement of α at $D25 \rightarrow D26$ and/or the specific bundle geometry assumed in that paper would be falsified.

A second, weaker prediction is that the WLM chain is *complete* in the sense of Definition 2(ii): no additional constant is required to stabilize any transition in the range D22–D27. This could in principle be tested by constructing explicit examples of dimensional transitions that fail all three stability conditions under any constant not in $\mathcal{C}$.

Relation to Existing Mathematical Frameworks

The WLM approach has structural affinities with several existing research programs. The treatment of constants as fixed points of functional equations is related to the theory of iterated function systems (Barnsley 1988) and to fixed-point theorems in functional analysis (Banach 1922). The use of the Atiyah-Singer index theorem to derive coupling constants has precedents in string theory and in noncommutative geometry (Connes 1994), where the standard model of particle physics is derived from spectral triples; within WLM this approach is developed specifically for α in Gu (2026a). The treatment of Ω as a structural boundary draws on algorithmic information theory (Li and Vitányi 2008). The key departure from all these precedents is the claim that the constants form a *chain* — a dependent sequence — rather than an independent collection, and that the chain is minimal and complete with respect to the dimensional transition structure.

Conclusion

The conventional view of mathematical constants treats them as numerical discoveries — facts about an independent mathematical reality that human investigation gradually uncovers. The WLM framework proposes a different reading: constants are structural invariants that are generated by — and in turn stabilize — specific types of recursive operation. They are not discovered; they are the fixed points of the generative processes that constitute mathematical reality itself.

This paper has formalized this reading for the constants $0, 1, \pi, e, \varphi, \alpha, \tau, \Omega, c$, and $\hbar$ in the context of the WLM dimensional transitions D22–D27. Full formal definitions have been provided for the structural space $\mathcal{S}$, the offset, rotation, and generative operators, the dimensional domains and their transition maps, and the four generative failure modes (divergence, collapse, desynchronization, incompressibility overflow). Proof sketches for the uniqueness of each constant as a stabilizer have been given; the topological derivation of α is treated separately in the companion paper (Gu 2026a), and the present paper summarizes only its structural role. The placement of Ω at the incompressibility boundary has been grounded in the definition of structural density and in the formal connection to Gödelian incompleteness.

The dependency graph of Section 12.2 makes precise the sense in which the chain is not a list but a coupled structure: each element load-bears for all subsequent ones. The open question of whether the geometric computation in Gu (2026a) extends to reproduce α^{-1} to full precision, alongside the extension of the chain beyond D27 into the Operator levels D28–D35, are the primary targets for future work.

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Mathematical Preliminaries

This appendix collects the foundational definitions needed for self-contained reading. Readers familiar with functional analysis, Kolmogorov complexity, and fiber bundle theory may skip to the relevant sections of the main text.

Fixed Points

Definition 23 (Fixed Point). Let $f: X \rightarrow X$ be a map on a set X . A point $x^* \in X$ is a fixed point of f if $f(x^*) = x^*$. The fixed-point set of f is $\text{Fix}(f) = \{x \in X: f(x) = x\}$.

Theorem 3 (Banach Fixed-Point Theorem). If (X, d) is a complete metric space and $f: X \rightarrow X$ is a contraction (i.e., $d(f(x), f(y)) \leq L \cdot d(x, y)$ for some $L < 1$), then f has a unique fixed point x^* , and the iterates $f^n(x)$ converge to x^* for any starting point $x \in X$.

Self-Similarity

Definition 24 (Self-Similar Set). A compact set $K \subset \mathbb{R}^n$ is self-similar with ratio $\lambda \in (0, 1)$ if it satisfies the Hutchinson equation: $K = \bigcup_{i=1}^m f_i(K)$, where each f_i is a contractive similarity with ratio λ . The Hausdorff dimension of K satisfies $m\lambda^d = 1$, giving $d = \log m / \log(1/\lambda)$.

Kolmogorov Complexity and Incompressibility

Definition 25 (Kolmogorov Complexity). *The Kolmogorov complexity of a binary string x with respect to a universal Turing machine U is: $K_U(x) = \min\{|p|: U(p) = x\}$, the length of the shortest program that outputs x . A string x of length n is incompressible (or Kolmogorov random) if $K_U(x) \geq n - c$ for some fixed constant c independent of x .*

Most strings are incompressible: the fraction of n -bit strings with $K(x) < n - k$ is at most 2^{-k} . However, specific constants such as π and e are *highly compressible* (their digits are generated by short programs). The Chaitin constant Ω is the canonical example of an incompressible real number.

Action and Propagation

Definition 26 (Action). *In classical mechanics, the action of a trajectory $q: [t_1, t_2] \rightarrow Q$ on a configuration space Q with Lagrangian L is: $S[q] = \int_{t_1}^{t_2} L(q(t), \dot{q}(t), t) dt$. In the quantum setting, the minimal action is $\hbar$, the reduced Planck constant, which sets the scale of quantum fluctuations via the path integral $\int \mathcal{D}q e^{iS[q]/\hbar}$.*

Definition 27 (Propagation Speed). *The propagation speed of a wave or signal in a medium is the rate at which a disturbance at one point influences a distant point. In special relativity, c is the universal upper bound on propagation speed, arising from the Lorentz-invariant spacetime geometry.*

Generative Operators

Definition 28 (One-Parameter Semigroup). *A strongly continuous one-parameter semigroup on a Banach space X is a family $\{T(t)\}_{t \geq 0}$ of bounded operators satisfying:*

1. $T(0) = \text{id}_X$.
2. $T(s + t) = T(s)T(t)$ for all $s, t \geq 0$.
3. $\lim_{t \rightarrow 0^+} T(t)x = x$ for all $x \in X$.

The generator of the semigroup is $A = \lim_{t \rightarrow 0^+} (T(t) - \text{id})/t$, defined on its domain $\mathcal{D}(A) \subset X$. The unique semigroup generated by $A = \lambda \cdot \text{id}$ is $T(t) = e^{\lambda t} \cdot \text{id}$, which is the WLM self-amplification model with base e^λ .

Structural Norms and Hilbert Spaces

Definition 29 (Hilbert Space). *A Hilbert space H is a complete inner product space: a vector space over $\mathbb{R}$ (or $\mathbb{C}$) equipped with an inner product $\langle \cdot, \cdot \rangle$ such that the induced norm $\|x\| = \sqrt{\langle x, x \rangle}$ makes H a complete metric space. Hilbert spaces support the spectral theorem for bounded self-adjoint operators, which is the technical basis for the eigenvalue arguments in Sections 4–11.*

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